GoLoG: Global-to-Local Decoupling Graph Network With Joint Optimization for Hyperspectral Image Classification

Bing Yang, Hailiang Ye, *Member, IEEE*, Ming Li, *Member, IEEE*, Feilong Cao, and Shirui Pan, *Senior Member, IEEE*

***Abstract*— Graph neural networks (GNNs) have a powerful ability to capture long-range spatial correlations in hyperspectral images (HSIs). However, existing GNN-based HSI classification methods are vulnerable to hand-crafted graphs, as the manner in which these graphs are constructed are often inappropriate and are likely to violate intrinsic graph properties, such as sparsity and low-rank. More importantly, the goal of HSI classification is to categorize each individual pixel into a land-cover class, but existing methods usually overuse global dependencies and ignore the importance of individualized spectral characteristics. Therefore, this article proposes a Global-to-Local decoupling Graph network (GoLoG) to conduct HSI classification in a global-to-local framework, which jointly optimizes the graph structure and network parameters guided by both intrinsic graph properties and classification loss. Specifically, a novel global-to-local network framework with successive global and local graph convolutional stages is constructed. By decoupling global and local stages, global contextual information can be exploited, and the individualized information of each hyperspectral pixel can be emphasized for HSI classification. Second, a sparse and low-rank graph structure learning model is proposed to refine and renovate the initial-construct graph. Finally, to unify graph structure learning and network training, a joint alternating update algorithm is introduced to jointly optimize the sparse and low-rank graph structure learning model and the global-to-local network framework. Extensive experiments demonstrate that the proposed GoLoG has obvious advantages compared with other state-of-the-art HSI classification methods.**

***Index Terms*— Graph convolutional network (GCN), graph structure learning, hyperspectral image (HSI) classification.**

Manuscript received 9 July 2023; revised 10 October 2023; accepted 15 October 2023. Date of publication 19 October 2023; date of current version 25 October 2023. This work was supported in part by the Zhejiang Provincial Natural Science Foundation of China under Grant LQ23F020012 and in part by the Fundamental Research Funds for the Provincial Universities of Zhejiang under Grant 2022YW65. The work of Ming Li was supported in part by the National Natural Science Foundation of China under Grant 62172370 and Grant U21A20473, and in part by the Zhejiang Provincial Natural Science Foundation under Grant LY22F020004. The work of Feilong Cao was supported by the National Natural Science Foundation of China under Grant 62176244. *(Corresponding author: Ming Li.)*

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Digital Object Identifier 10.1109/TGRS.2023.3325459

I. Introduction

HYPERSPECTRAL image (HSI) classification is a hot topic in the field of remote sensing. Given a set of observations (i.e., the hyperspectral pixels in an HSI), HSI classification aims to categorize each pixel into one of several land-cover classes. Abundant spectral and spatial distribution information of land covers help to reveal the intrinsic structure of Earth observation and to precisely distinguish different physical materials, which makes HSI classification a competitive technology for many application scenarios, such as precision agriculture, smart city, and environmental surveillance [1], [2], [3], [4].

In recent years, deep learning has made significant breakthroughs in the task of HSI classification [5], [6], [7], [8], [9], [10]. Particularly, convolutional neural networks (CNNs), which are representative networks with a powerful spatial feature extraction ability, have achieved unprecedented advances and become the fundamental infrastructure of state-of-the-art methods [11], [12], [13], [14], [15]. For example, Yu et al. [16] proposed a simple but efficient 2-D-CNN model to extract valuable features for accurate HSI classification. Cao et al. [17] integrated Markov random fields and CNNs to make full use of the spatial information of an HSI and further improve classification performance. Gao et al. [14] devised a mixed depthwise convolution-based multiscale residual network to extract deep features with fewer network parameters. Zhong et al. [18] built a spectral–spatial residual network (SSRN) with consecutive spectral and spatial residual blocks to obtain discriminative spectral–spatial features, which greatly improves classification accuracy. Subsequently, Sun et al. [19] designed a spectral–spatial attention network to suppress the effects of interfering pixels and further extract effective deep features from the attention areas of HSI. In [20], a spatial–spectral attention network based on 3-D octave convolution is presented to extract discriminative features for HSI classification. In [21], a novel feature-grouped network (FGN) was designed with the aim to slice HSI data into several groups along the spectral dimension and then learn complementary spectral–spatial features from them. Additionally, Wang et al. [22] proposed a lightweight multiscale attention fusion network (LMAFN) to explore the complementary information of different hierarchical structures in HSI and simultaneously achieve a lower computation cost. Nevertheless, the regular

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convolution kernel and limited receptive field greatly restrict the CNN in advanced HSI feature extraction, as different objects always possess various geometric characteristics, such as shapes and sizes.

Graph neural networks (GNNs) have received significant attention owing to their exceptional capacity for long-range modeling of HSIs [23], [24], [25], [26]. In particular, many methods based on graph convolutional networks (GCNs) have been developed and have shown superiority in HSI classification. For instance, Hong et al. [27] developed a minibatch GCN to capture the long-range spatial information of HSI and achieved satisfactory classification performance. In [28], a dual GCN is presented to improve classification accuracy under limited training samples. Instead of working on pixel-based graphs, most methods propose to segment the HSI into superpixels and then perform on superpixel-based graphs to reduce the computational cost. For example, Yu et al. [29] designed a two-branch deeper GCN to make use of the advantages of GCN and CNN, which achieves better classification performance. Zhou et al. [30] built two subnetworks of multiscale fully CNN and multihop GCN, and then combined them to obtain an attention multihop graph and multiscale convolutional fusion network (AMGCFN) to exploit the multilevel information of HSI. Although promising results have been achieved, these GCN-based methods are vulnerable to the hand-crafted graphs, and the classification performance could greatly degrade with an inappropriate hand-crafted graph.

Recent works have demonstrated that adaptively learning the accurate and suitable graph structure of HSI is of vital importance for improving HSI classification accuracy. For instance, Wan et al. [31] presented a multiscale dynamic GCN (MDGCN) by constructing multiple graphs with different scales to extract multiscale spatial information, where these graphs can be dynamically updated during the graph convolution process. Liu et al. [32] built a CNN-enhanced GCN (CEGCN) to exploit both pixel-level and superpixel-level features, where the edge weights of graphs can be adaptively learned by calculating the similarities between hidden features. Wan et al. [33] developed a similarity measurement to refine the connective relationships and edge weights of graphs, and the obtained model can adaptively exploit contextual relations. Ding et al. [34] utilized graph topological consistent connectivity to adaptively explore global high-order neighbors and further capture rich spatial contextual information. Bai et al. [35] calculated the cosine similarities between output features and then update the graph structure to improve classification performance. Chen et al. [36] embedded a semisupervised Siamese network into GCN to dynamically update graph structure and further accomplish automatic graph learning. Yang et al. [37] learned dynamic graphs across different layers and then adaptively integrated them with the initial graph to extract robust features from HSI. These methods can adaptively learn the graph structure from the HSI by means of neural networks or similarity learning, and thus boost classification performance.

However, several important issues still exist with these GCN-based HSI classification methods. First, it is essential to recognize that real-world graphs are inherently characterized by specific attributes, notably sparsity and low-rank properties. However, we observe a discrepancy where prevailing graph-learning approaches tend to compromise or overlook these intrinsic characteristics, resulting in potential misalignments with the fundamental nature of real-world graph structures. Second, to guarantee the long-range modeling ability, most existing approaches typically learn graph structure based on a pairwise similarity between hyperspectral pixels, and then aggregate information guided by the obtained graph. However, this strategy has a potential drawback. Although it is helpful to resort to hyperspectral pixels that are far away in space but belong to the same class, the overuse of global dependencies can easily weaken individualized spectral information, which is unfavorable for HSI classification whose goal is to categorize each pixel into a land-cover class. Third, as the purpose of learning a graph structure is also to improve classification performance, it is very important for GCN-based HSI classification methods to integrate graph structure learning and network parameter training. However, most existing methods always isolate the processes of learning a graph structure and network parameters, which could lead to a suboptimal graph structure.

To address the above issues, this article proposes a global-to-local decoupling graph network (GoLoG) in an attempt to refine the utilization of global dependency and local dependency for high-quality HSI classification. Specifically, considering the different importance of global and local information, a novel global-to-local decoupling framework with successive global and local graph convolutional stages is constructed to first mine global dependencies and then emphasize local individualized correlations. The global graph convolutional stage aims to aggregate the long-range dependency information, while the local graph convolutional stage focuses on collecting the neighborhood dependency information. In this way, the proposed framework can effectively adapt to the HSI classification. Second, each HSI's superpixel generally relies only on a finite number of nodes, and superpixel nodes representing the same material in HSIs often have similar neighbor nodes. The sparsity and approximate low-rank nature are expressed through the respective manifestations of these inherent properties within the graph structure. In response, we introduce a novel learning model that is tailored to accommodate the sparse and low-rank characteristics intrinsic to such structures, aiming to refine the initial graph and amplify the distinctiveness of the extracted feature representations. This model is particularly characterized by the integration of sparse and low-rank constraints on the graph being learned, a strategy grounded in the preservation of these fundamental properties. The incorporation of these constraints facilitates a transformation into a learning model that is both structurally and functionally aligned with the sparse and low-rank nature of graph data. Finally, as the joint optimization of the graph structure and network framework is important but

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challenging, this article introduces an effective joint alternating update algorithm to iteratively update the graph structure and network parameters until the graph and parameters achieve satisfactory performance.

The proposed GoLoG distinguishes itself from the existing methods by exceling in three dimensions. First, we highlight the distinct roles of global and local information within the context of HSI classification, employing a refined approach that integrates these elements effectively within our advanced global-to-local network framework. This approach stands in contrast to existing studies, which often overlook the diverse impacts of these distinct informational paradigms. Second, GoLoG advances a novel sparse and low-rank graph structure learning model, carefully crafted to enhance the initial graph, yielding a refined structure that intrinsically aligns with sparse and low-rank attributes. This progression is not typically observed in conventional approaches, where the learning of graph structures is often anchored in neural networks or similarity learning, with a noticeable absence of considerations for these inherent graph characteristics. Third, and perhaps most significantly, GoLoG incorporates a joint alternating optimization method, facilitating the simultaneous optimization of both graph structure and associated parameters. This integrated approach markedly contrasts with the normative paradigms where graph learning and parameter training are often distinctly isolated, a separation that potentially undermines the holistic optimization of the learning process.

The main contributions are summarized as follows.

1) A novel GoLoG is proposed to mine the global dependencies and emphasize the local individualized correlations for highly accurate HSI classification.
2) A sparse and low-rank graph structure learning model guided by intrinsic prior graph properties is developed to renovate the graph structure. The refined graph helps to extract significantly more reliable and well-directed graph representations of HSI.
3) A joint alternating update algorithm is presented to jointly optimize the graph structure and the network parameters, which unifies the processes of graph structure learning and parameter training, and further significantly boosts the HSI classification accuracy.

The remainder of this article is organized as follows. Section II gives some notations and introduces GCN. Section III elaborates the proposed GoLoG. Section IV presents the experiment results on several real HSI datasets. Section V concludes this article.

II. Preliminary

A. Notations

This section defines some notations used in this article. Let $\boldsymbol{Z} \in \mathbb{R}^{H\times W\times B}$ denote an HSI, where H, W, and B are the height, the width, and the number of spectral bands of $\boldsymbol{Z}$, respectively. Assume that there are C classes in the HSI $\boldsymbol{Z}$, and the set of class labels is denoted by $\mathcal{C} = \{1, 2, \ldots, C\}$. In this work, $\mathcal{D}_L = \{(\boldsymbol{z}_{l_i}, y_i)\}_{i=1}^{N}$ denotes the labeled set with N labeled samples, here $\boldsymbol{z}_{l_i} \in \mathbb{R}^B$ and $y_i \in \mathcal{C}$ are the spectral signature and the class label of the ith labeled sample, respectively. Moreover, the Frobenius norm, the ℓ_1 norm, and the nuclear norm of a matrix $\boldsymbol{E} = (E_{ij})_{h\times w}$ are, respectively, defined as $\|\boldsymbol{E}\|_F^2 = \sum_{ij} E_{ij}^2$, $\|\boldsymbol{E}\|_1 = \sum_{ij} |E_{ij}|$, and $\|\boldsymbol{E}\|_* = \sum_{i=1}^{\min\{h,w\}} \sigma_i$, where $\sigma_i \geq 0$ is the ith singular value of $\boldsymbol{E}$. $(\boldsymbol{E})_+$ represents the elementwise positive part of $\boldsymbol{E}$, i.e., $E_{ij} = \max\{E_{ij}, 0\}$.

B. Graph Convolutional Network

As a prominent example of GNN, GCNs have shown powerful capability in exploiting complex interactions among data [38], [39], [40]. Given a graph with K nodes, and let $\boldsymbol{X}^{\text{in}} \in \mathbb{R}^{K\times d}$ represents the feature matrix where each row denotes a d-dimension feature for corresponding node in this graph. Then, a typical type of graph convolution [41] can be defined as

$$\boldsymbol{X}^{\text{out}} = \phi\left(\tilde{\boldsymbol{A}}^{*}\boldsymbol{X}^{\text{in}}\boldsymbol{W}\right) \tag{1}$$

where $\boldsymbol{W} \in \mathbb{R}^{d\times f}$ is a learnable weight matrix, $\phi(\cdot)$ is a nonlinear activation function, and $\tilde{\boldsymbol{A}}^{*}$ is the normalized adjacency matrix of this graph with $\tilde{\boldsymbol{A}}^{*} = \tilde{\boldsymbol{D}}^{-1/2}(\tilde{\boldsymbol{A}} + \boldsymbol{I}_K)\tilde{\boldsymbol{D}}^{-1/2} \in \mathbb{R}^{K\times K}$, where $\tilde{\boldsymbol{A}} = (\tilde{A}_{ij})$ is the adjacency matrix, $\boldsymbol{I}_K$ denotes the identity matrix, and $\tilde{\boldsymbol{D}} = (\tilde{D}_{ii})$ is the degree matrix of $\tilde{\boldsymbol{A}} + \boldsymbol{I}_K$ with the diagonal element $\tilde{D}_{ii} = \sum_j (\tilde{A}_{ij} + I_{ij})$.

III. Global-to-Local Decoupling Graph Network

A complete overview of GoLoG is given in this section. As shown in Fig. 1, the GoLoG conducts HSI classification in a global-to-local network framework with the help of a jointly optimizing graph structure and network framework. Specifically, the HSI is first segmented into a large number of superpixels. Then, a global-to-local network framework with successive global and local graph convolutional stages is utilized to fully mine the global information and local individualized information of HSI. In the global graph convolutional stage, a global graph is constructed according to spectral similarities between superpixels, and a global graph convolutional module (GGCM) is built to aggregate the long-range dependency information. In the local graph convolutional stage, a local graph is constructed based on spatial adjacency relations among superpixels, and a local graph convolutional module (LGCM) is proposed to collect the neighborhood dependency information. Furthermore, in the global graph convolutional stage, a sparse and low-rank graph structure learning model is proposed to refine the initial global graph. Finally, a joint alternating update algorithm is introduced to unify and jointly optimize the sparse and low-rank graph structure learning model and the global-to-local network framework.

Before continuing, we need to do superpixel segmentation for HSIs to exploit the global information and local individualized information of HSI. To this end, simple linear iterative clustering (SLIC) [42] is first utilized to segment the HSI into $M = \lceil (H \times W)/\mu \rceil$ superpixels, where $\mu \geq 1$ controls the segmentation scale. The spectral signature of each superpixel, denoted by $\boldsymbol{x}_i \in \mathbb{R}^B$, is obtained by averaging the spectral signatures of all pixels in this superpixel. Then, these M

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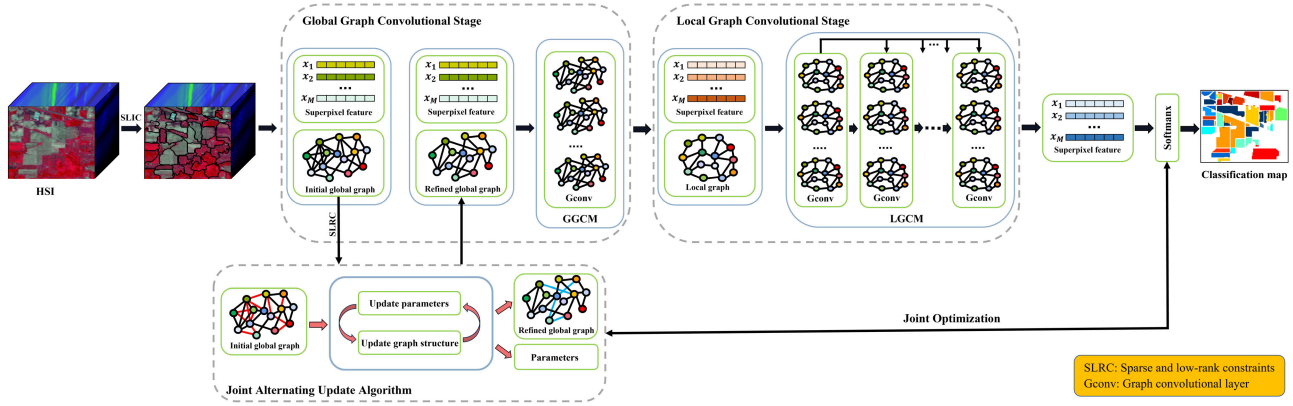


Fig. 1. Overall framework of the proposed GoLoG. First, the HSI is segmented into a large number of superpixels. Then, the GGCM is used to aggregate the global dependency information, where the global graph can be refined by the sparse and low-rank graph structure learning model. Next, the LGCM is utilized to mine the neighborhood dependency information. Finally, the joint alternating update algorithm is used to jointly optimize the graph structure learning model and the network framework.

superpixels are used as the nodes of the global graph and local graph, and the corresponding node feature matrix is denoted by $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_M]^T \in \mathbb{R}^{M \times B}$.

Next, we will present the proposed method in detail.

A. Global Graph Convolutional Stage

This section proposes a global-to-local network framework with successive global and local graph convolutional stages to conduct the HSI classification task. The antecedent global graph convolutional stage mines complementary information from the same-class hyperspectral pixels that are far away in space, while the following local graph convolutional stage greatly enhances the local individualized information to better distinguish hyperspectral pixels and further classify them correctly.

In the global graph convolutional stage, an initial global graph is first constructed based on the pairwise similarity between superpixel nodes, to acquire the long-range dependencies within the whole HSI. Then, the initially constructed global graph is refined by the proposed sparse and low-rank graph structure learning model. In the end, the GGCM is proposed to aggregate information guided by the refined global graph, which greatly helps to maintain the smoothness of large homogeneous regions in HSI. The concrete process can be described as the following three phases.

1) Global Graph Construction: The global graph is obtained by using the cosine similarity between the spectral signatures of each superpixel node pair in HSI. To be more precise, we first calculate the cosine similarities of all node pairs as

$$S_{ij} = \frac{\boldsymbol{x}_i{}^T \boldsymbol{x}_j}{\|\boldsymbol{x}_i\| \|\boldsymbol{x}_j\|} \quad \forall i, j = 1, 2, \ldots, M. \tag{2}$$

Then, for each superpixel node, the other superpixels with the largest k similarities are selected as its neighbors, and the value of similarity is used as the weight of the corresponding edge in the global graph. Specifically, for ith superpixel node, the kth largest similarity of ith row in $\boldsymbol{S}$ is selected as the similarity threshold $\tau_i (i = 1, 2, \ldots, M)$, and then the adjacency matrix of the global graph, denoted as $\boldsymbol{A}_g$, can be obtained as

$$(\boldsymbol{A}_g)_{ij} = \begin{cases} S_{ij}, & \text{if } S_{ij} \geq \tau_i \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where $i, j = 1, 2, \ldots, M$.

However, the initially constructed global graph based on pairwise similarity is always noisy and contains many wrong edges, as the complicated spectral variability in HSI means the spectral signatures are not able to always accurately express common characteristics of same-class pixels, especially those pixels that are far away in space. This will inevitably weaken the discrimination of the extracted feature representations. Furthermore, the intrinsic graph properties, namely sparsity and low-rank, are not considered during the initially constructed process of the global graph. Therefore, the initially constructed global graph is refined and updated.

2) Sparse and Low-Rank Graph Structure Learning Model: Given the initially constructed global graph $\boldsymbol{A}_g$, the global graph refinement process with sparse and low-rank constraints can be formulated as

$$\mathcal{L}_{\text{SLR}}(\boldsymbol{A}) = \left\|\boldsymbol{A}_g - \boldsymbol{A}\right\|_F^2 + \alpha \|\boldsymbol{A}\|_0 + \beta \text{rank}(\boldsymbol{A}) \tag{4}$$

where $\alpha > 0$ and $\beta > 0$ are two hyperparameters; $\|\boldsymbol{A}\|_0$ represents the number of nonzero elements of $\boldsymbol{A}$ to describe its sparsity; $\text{rank}(\boldsymbol{A})$ is the number of nonzero singular values of $\boldsymbol{A}$, which enforces its low-rank property; $\|\boldsymbol{A}_g - \boldsymbol{A}\|_F^2$ guarantees that the refined adjacency matrix $\boldsymbol{A}$ should be close to $\boldsymbol{A}_g$. It has been shown in [43], [44], and [45] that the ℓ_1-norm and the nuclear norm of a matrix is one feasible choice of sparse and low-rank, respectively. As a result, (4) can be rewritten as

$$\mathcal{L}_{\text{SLR}}(\boldsymbol{A}) = \left\|\boldsymbol{A}_g - \boldsymbol{A}\right\|_F^2 + \alpha \|\boldsymbol{A}\|_1 + \beta \|\boldsymbol{A}\|_* \tag{5}$$

where the term $\|\boldsymbol{A}\|_1$ is the ℓ_1-norm, which ensures that the sparsity of the learned graph structure can be retained since each HSI's superpixel generally depends on a finite number of nodes. Further, theoretically, superpixel nodes representing the same material in HSIs often have similar neighbor nodes. This can be reflected as the approximate low-rank property of the adjacency matrix. Therefore, we adopt the nuclear norm term $\|\boldsymbol{A}\|_*$ to represent this property.

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The reconstructed and refined sparse and low-rank global graph $\boldsymbol{A}$ is utilized to appropriately guide the process of aggregating the global spatial information of HSI. The detailed optimization process is introduced in Section III-D. The normalized adjacency matrix of $\boldsymbol{A}$ is defined as

$$\tilde{\boldsymbol{A}} = \tilde{\boldsymbol{D}}^{-1/2}(\boldsymbol{A} + \boldsymbol{I}_M)\tilde{\boldsymbol{D}}^{-1/2} \tag{6}$$

where $\tilde{\boldsymbol{D}}$ is the degree matrix of $\boldsymbol{A} + \boldsymbol{I}_M$.

3) Global Graph Convolutional Module: Given the refined global graph, the GGCM can be typically formulated as

$$\boldsymbol{H}_g = \varphi\left(\tilde{\boldsymbol{A}}\boldsymbol{H}_{\text{pre}}\boldsymbol{W}_g\right) \tag{7}$$

where $\boldsymbol{W}_g \in \mathbb{R}^{n\times n}$ is the learnable weight matrix, n denotes the feature dimension, and $\varphi(\cdot)$ represents the rectified linear unit (ReLU). Before being fed into the GGCM, the input $\boldsymbol{X}$ is first converted to a n-dimension feature $\boldsymbol{H}_{\text{pre}}$ through a fully connected layer, i.e., $\boldsymbol{H}_{\text{pre}} = \boldsymbol{X}\boldsymbol{W}_{\text{pre}}$, where $\boldsymbol{W}_{\text{pre}} \in \mathbb{R}^{B\times n}$ is a learnable weight matrix.

B. Local Graph Convolutional Stage

After acquiring and utilizing long-range dependencies within the whole HSI, the important information among local spatial regions is captured and enhanced. Specifically, a local graph structure is first constructed according to the spatial adjacency relations. Then, several local graph convolutions are conducted to make full use of the local individualized information contained in the local graph, which is important to distinguish pixels and further improve classification performance. The details are given as follows.

1) Local Graph Construction: Let $\boldsymbol{A}_l \in \mathbb{R}^{M\times M}$ be the adjacency matrix of the local graph, whose element represents the spatial adjacency relation and the degree of correlation between each pair of superpixel nodes and can be calculated as

$$(\boldsymbol{A}_l)_{ij} = \begin{cases} \exp\left(-\nu\left\|\boldsymbol{x}_i - \boldsymbol{x}_j\right\|^2\right), & \text{if nodes } i \text{ and } j \\ & \text{are adjacent} \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

where superpixel nodes i and j are adjacent which means that there exists at least one common boundary between them, and $\nu = 0.2$ is a hyperparameter. By considering the relations between locally spatial adjacent superpixel nodes, $\boldsymbol{A}_l$ reflects the local spatial information of the HSI. Then, the normalized adjacency matrix of $\boldsymbol{A}_l$ can be given as

$$\tilde{\boldsymbol{A}}_l = \tilde{\boldsymbol{D}}_l^{-1/2}(\boldsymbol{A}_l + \boldsymbol{I}_M)\tilde{\boldsymbol{D}}_l^{-1/2} \tag{9}$$

where $\tilde{\boldsymbol{D}}_l$ represents the degree matrix of $\boldsymbol{A}_l + \boldsymbol{I}_M$.

In the local graph convolutional stage, the local graph is directly used to guide information aggregation without graph structure learning. The reason why the local graph is not learned is that the spatial adjacency relation is trustworthy enough to reveal the local spatial information. HSIs have lots of homogeneous regions, and neighboring superpixels within the same homogeneous region are more likely to belong to the same-class material and should be connected.

2) Local Graph Convolutional Module: To fully exploit local individualized information, the LGCM should contain several local graph convolutions. Nevertheless, simply stacking multiple conventional graph convolutional layers will face the over-smoothing problem and result in performance degradation. To alleviate this problem, the initial residual connection and the identity mapping [46] are integrated into the LGCM to ensure the separability of the extracted features in higher graph convolutional layers. Specifically, the residual connection to the initial features is used to guarantee that the extracted features of superpixel nodes include some initial spectral information without becoming indistinguishable. Identity mapping is incorporated into the weight matrix to ensure that the LGCM at least achieves comparable performance to its shallow variant.

In summary, assume there are Q local graph convolutional layers in the LGCM, then the mathematical expression of the qth layer can be described as

$$\boldsymbol{H}^{(q)} = \varphi\left(\left((1-\rho)\tilde{\boldsymbol{A}}_l\boldsymbol{H}^{(q-1)} + \rho\boldsymbol{H}^{(0)}\right)\left((1-\vartheta)\boldsymbol{I}_n + \vartheta\boldsymbol{W}_l^{(q)}\right)\right) \tag{10}$$

where $\boldsymbol{W}_l^{(q)} \in \mathbb{R}^{n\times n}$ is the learnable weight matrix, $\rho \in [0, 1]$ and $\vartheta \in [0, 1]$ are two hyperparameters, and $q = 1, 2, \ldots, Q$. The features extracted by the GGCM are used as the input of the LGCM, i.e., $\boldsymbol{H}^{(0)} = \boldsymbol{H}_g$.

C. Joint Loss of GoLoG

The HSI classification aims to assign a class label to each hyperspectral pixel. Thus, the learned features of these superpixels need to be transformed back to original HSI space. That is, the learned feature of the superpixel node will be assigned to pixels who belong to this superpixel. After this process, the features of all hyperspectral pixels can be obtained. Mathematically, assume that $\boldsymbol{Z}^*$ represents the learned features of the HSI and $\mathcal{D}_L^* = \{(\boldsymbol{z}_{l_i}^*, y_i)\}_{i=1}^N$ denotes the corresponding labeled set. Let $\hat{y}_i = \text{softmax}(\boldsymbol{W}_c^T\boldsymbol{z}_{l_i}^*)$ be the predicted label of ith samples, where $\boldsymbol{W}_c \in \mathbb{R}^n$ is the weight matrix in the classification layer and softmax$(\cdot)$ represents the softmax function. Then, the classification loss of HSIs is defined as

$$\mathcal{L}_{\text{class}}(\boldsymbol{W}, \boldsymbol{A}) = \frac{1}{\left|\mathcal{D}_L^*\right|}\sum_{\left(\boldsymbol{z}_{l_i}^*, y_i\right)\in\mathcal{D}_L^*} \ell\left(\hat{y}_i, y_i\right) \tag{11}$$

where $\boldsymbol{W} = \{\boldsymbol{W}_g, \boldsymbol{W}_{\text{pre}}, \{\boldsymbol{W}_l^{(q)}\}_{q=1}^Q, \boldsymbol{W}_c\}$ denotes all the parameters of GoLoG, $\boldsymbol{A}$ is the learnable global graph, and $\ell(\cdot,\cdot)$ represents the cross entropy loss [47].

Intuitively, the renovated and refined sparse-low-rank global graph $\boldsymbol{A}$ can be obtained in advance by directly optimizing (5), and then the classification model can be trained based on the learned global graph. However, as previously mentioned, this preprocessing strategy will isolate the processes of learning the graph structure and network parameters, and the global graph learned by such two-stage strategy may be suboptimal for the classification model.

Thus, it is necessary to develop a better approach that can jointly optimize the graph structure and the classification

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model. To this end, a joint objective function constituted by both graph-structure-learning loss and classification loss is built as

$$\begin{aligned}\min_{\boldsymbol{W},\boldsymbol{A}} \mathcal{L}(\boldsymbol{W},\boldsymbol{A}) &= \mathcal{L}_{\text{class}}(\boldsymbol{W},\boldsymbol{A}) + \mathcal{L}_{\text{SLR}}(\boldsymbol{A}) \\ &= \mathcal{L}_{\text{class}}(\boldsymbol{W},\boldsymbol{A}) + \left\|\boldsymbol{A}_g - \boldsymbol{A}\right\|_F^2 + \alpha\|\boldsymbol{A}\|_1 \\ &\quad + \beta\|\boldsymbol{A}\|_*. \end{aligned} \tag{12}$$

A potential benefit of this joint loss is that $\mathcal{L}_{\text{class}}(\boldsymbol{W},\boldsymbol{A})$ can also guide the process of graph structure learning to refine an inappropriate global graph and further improve the classification performance.

D. Joint Alternating Update Algorithm

It is challenging to jointly optimize $\boldsymbol{W}$ and $\boldsymbol{A}$ in (12) due to the sparse and low-rank constraints on $\boldsymbol{A}$. Therefore, an alternate iterative optimization strategy is used to update $\boldsymbol{W}$ and $\boldsymbol{A}$, i.e., at the tth ($t = 1, 2, \ldots, T$) step, we update the solution by fixing one variable and solving the remaining one.

1) Update $\boldsymbol{W}$: For (12), ignoring constant terms, we can use the stochastic gradient descent (SGD) algorithm to update $\boldsymbol{W}$ as

$$\boldsymbol{W}^{(t)} = \boldsymbol{W}^{(t-1)} - \eta \frac{\partial \mathcal{L}_{\text{class}}\left(\boldsymbol{W}, \boldsymbol{A}^{(t-1)}\right)}{\partial \boldsymbol{W}}\bigg|_{\boldsymbol{W}=\boldsymbol{W}^{(t-1)}} \tag{13}$$

where $\boldsymbol{W}^{(t-1)}$ and $\boldsymbol{A}^{(t-1)}$ are, respectively, the $(t-1)$th solution of $\boldsymbol{W}$ and $\boldsymbol{A}$.

2) Update $\boldsymbol{A}$: When updating $\boldsymbol{A}$, $\boldsymbol{W}^{(t)}$ is fixed while the other terms that are irrelevant to $\boldsymbol{A}$ are removed, and the objective of (12) arrives at

$$\mathcal{L}\left(\boldsymbol{W}^{(t)}, \boldsymbol{A}\right) = \hat{\mathcal{L}}\left(\boldsymbol{A}, \boldsymbol{A}_g\right) + \alpha\|\boldsymbol{A}\|_1 + \beta\|\boldsymbol{A}\|_* \tag{14}$$

where

$$\hat{\mathcal{L}}\left(\boldsymbol{A}, \boldsymbol{A}_g\right) = \mathcal{L}_{\text{class}}\left(\boldsymbol{W}^{(t)}, \boldsymbol{A}\right) + \|\boldsymbol{A}_g - \boldsymbol{A}\|_F^2. \tag{15}$$

It can be found that (14) is split into two parts, where the first part $\hat{\mathcal{L}}(\boldsymbol{A}, \boldsymbol{A}_g)$ is differentiable while $\|\boldsymbol{A}\|_1$ and $\|\boldsymbol{A}\|_*$ are nondifferentiable. This objective function with two nondifferentiable regularizers in (14) can be optimized by proximal splitting method [48], formulated as

$$\begin{cases} \boldsymbol{A}^{(t)} = \boldsymbol{A}^{(t-1)} - \eta \dfrac{\partial \hat{\mathcal{L}}\left(\boldsymbol{A}, \boldsymbol{A}_g\right)}{\partial \boldsymbol{A}}\bigg|_{\boldsymbol{A}=\boldsymbol{A}^{(t-1)}} \\ \boldsymbol{A}^{(t)} = \text{prox}_{\alpha\|\cdot\|_1}\left(\boldsymbol{A}^{(t)}\right) \\ \boldsymbol{A}^{(t)} = \text{prox}_{\beta\|\cdot\|_*}\left(\boldsymbol{A}^{(t)}\right) \end{cases} \tag{16}$$

where $\text{prox}_{\alpha\|\cdot\|_1}(\boldsymbol{P}) = \text{sgn}(\boldsymbol{P}) \odot (\boldsymbol{P} - \alpha)_+$, $\text{prox}_{\beta\|\cdot\|_*}(\boldsymbol{P}) = \boldsymbol{U}\text{diag}((\sigma_1 - \beta)_+, (\sigma_2 - \beta)_+, \ldots, (\sigma_M - \beta)_+)\boldsymbol{V}^T$, $\boldsymbol{P} = \boldsymbol{U}\text{diag}(\sigma_1, \sigma_2, \ldots, \sigma_M)\boldsymbol{V}^T$ is the singular value decomposition, and $\odot$ denotes Hadamard product of matrices.

To sum up, details of the joint alternating update algorithm for GoLoG are shown in Algorithm 1.

Algorithm 1 Joint Alternating Update Algorithm for GoLoG

Input: $\boldsymbol{X}$, $\boldsymbol{A}_g$, $\boldsymbol{A}_l$, hyperparameters α, β, ρ and ϑ, learning rate η, maximal iterations T.
Output: $\boldsymbol{A}$ and $\boldsymbol{W}$.

```
Set t = 1, A^(0) = A_g, and randomly initialize W^(0);
while t ≤ T do
    Compute H^(L) by Eq.(7) and Eq.(10);
    Convert H^(L) into the HSI space and obtain Z*;
    Compute L_class(W^(t-1), A^(t-1)) by Eq.(11);
    W^(t) = W^(t-1) − η ∂L_class(W, A^(t-1))/∂W |_{W=W^(t-1)};
    A^(t) = A^(t-1) − η ∂L̂(A, A_g)/∂A |_{A=A^(t-1)};
    A^(t) = prox_{α‖·‖_1}(A^(t));
    A^(t) = prox_{β‖·‖_*}(A^(t));
    t ← t + 1;
end
A ← A^(T).
Return: A and W.
```

IV. Experiment

A. Experiment Settings

1) Datasets: Three benchmark HSI datasets are utilized in the experiments, namely two datasets collected by the AVIRIS instrument, i.e., Salinas and Indian Pines, and one dataset acquired by ROSIS, i.e., Pavia University. The Salinas dataset comprises 512 × 214 pixels with 204 spectral bands after removing 20 bands that are polluted by water absorption, and the spatial resolution is 3.7 m per pixel. There are 16 different classes in the Salinas dataset. The Indian Pines dataset contains 200 spectral bands with a spatial size of 145×145 and a spatial resolution of 20 m, and the ground truth includes 16 different classes. The Pavia University dataset has 103 bands with 610×340 pixels and contains nine classes of urban land covers, and the spatial resolution of the Pavia University dataset is 1.3 m per pixel.

2) Comparison Methods: To verify its effectiveness, the GoLoG is compared against several state-of-the-art HSI classification models, including CNN-based methods, namely 2-D-CNN [16], SSRN [18], LMAFN [22], FGN [21], and also GCN-based methods, namely GCN, MDGCN [31], CEGCN [32], and AMGCFN [30]. It should be noted that all the GCN-based methods use the same superpixels as network inputs, and do not employ the CNN branch. The default hyperparameter settings of all the compared models are adopted. In addition to representative baselines, the GoLoG is also compared with its variant model, denoted by $\text{GoLoG}_{\text{base}}$. Specifically, $\text{GoLoG}_{\text{base}}$ consists of successive global graph convolutional stages and local graph convolutional stages but without exploring the sparse and low-rank properties of the global graph. The only difference between $\text{GoLoG}_{\text{base}}$ and GoLoG is that $\text{GoLoG}_{\text{base}}$ does not learn the global graph structure.

3) Parameter Setting: In the experiments, we set the hyperparameter α as 0.3, 0.8, and 0.7 for the Salinas, Indian Pines, and Pavia University datasets, respectively. β is, respectively, set as 0.3, 0.5, and 0.4, while Q is set as 6, 4, and 3,

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TABLE I
CLASSIFICATION ACCURACIES OF THE COMPARED METHODS ON THE SALINAS DATASET

Class	train	test	2D-CNN	SSRN	LMAFN	FGN	GCN	MDGCN	CEGCN	AMGCFN	$\text{GoLoG}_{\text{base}}$	GoLoG
1	11	1998	99.96	98.48	99.05	99.01	99.68	**100.00**	99.71	**100.00**	**100.00**	98.56
2	19	3707	98.81	**100.00**	98.44	**100.00**	99.79	99.16	**100.00**	99.99	**100.00**	99.88
3	10	1966	97.64	97.92	96.83	98.47	95.82	98.51	**100.00**	99.58	98.04	**100.00**
4	7	1387	97.52	97.90	97.18	98.52	93.48	92.41	95.48	98.96	**99.70**	99.64
5	14	2664	99.20	**99.83**	98.15	99.52	96.79	97.71	96.91	94.62	96.91	97.72
6	20	3939	99.80	99.93	99.86	99.60	98.86	99.39	99.23	**99.99**	99.81	99.57
7	18	3561	99.49	99.96	99.90	99.95	99.17	99.30	99.85	99.95	99.92	**99.96**
8	57	11214	89.12	88.81	85.85	87.62	94.46	97.75	97.27	97.56	98.90	**99.62**
9	32	6171	99.99	99.94	99.80	99.93	99.90	99.77	99.99	99.97	**100.00**	99.97
10	17	3261	92.57	96.23	94.87	93.96	91.57	91.29	95.07	**99.05**	92.55	96.98
11	6	1062	96.92	98.33	95.79	97.79	91.55	97.55	98.70	95.88	**98.79**	97.93
12	10	1917	99.66	100.00	99.44	100.00	99.82	99.97	98.16	98.55	**100.00**	99.98
13	5	911	99.85	99.70	**99.86**	99.70	97.12	98.13	93.24	94.87	98.20	97.95
14	6	1064	95.54	95.86	**98.21**	96.16	96.35	96.01	97.65	95.57	96.69	97.2
15	37	7231	91.46	87.65	77.39	86.13	96.80	98.75	99.37	96.64	99.34	**99.50**
16	10	1797	94.84	98.98	97.82	99.21	95.43	94.89	99.10	**100.00**	99.17	99.92
OA(%)	-	-	95.48±0.65	95.42±0.43	93.03±0.47	94.85±0.75	96.80±0.63	97.97±0.26	98.42±0.60	98.36±0.74	98.83±0.14	**99.32±0.23**
AA(%)	-	-	97.02±0.66	97.47±0.33	96.15±0.67	97.22±0.51	96.66±0.86	97.54±0.58	98.11±0.61	98.20±0.71	98.63±0.27	**99.02±0.40**
$\kappa \times 100$	-	-	94.97±0.73	94.90±0.48	92.24±0.52	94.27±0.83	96.44±0.71	97.74±0.29	98.24±0.67	98.17±0.82	98.70±0.16	**99.24±0.25**

respectively. Again, the number of hidden units is set to 128 for all three datasets. Following our previous work [37], ρ is, respectively, set as 0.3, 0.3, and 0.4, while ϑ is set as 0.5 for all three datasets. μ is set as 80, 25, and 80, respectively. k is set as 50 for the three datasets. Furthermore, the learning rate η is set to 0.001, and the number of training iterations T is 300. The key hyperparameters are discussed in Section IV-D. For the Salinas and Pavia University datasets, 0.5% of the labeled samples are randomly chosen as the training data, and the remaining are used as the testing data. For the Indian Pines dataset, 5% of the labeled samples are randomly chosen for training, and the remaining are utilized for testing. All experiments are conducted on a desktop computer with Intel Xeon W-2225 CPU and NVIDIA GeForce RTX 2080 Ti.

4) Indices: Overall accuracy (OA), average accuracy (AA), kappa coefficient (κ), and individual class accuracy are the performance indices used for the quantitative evaluation [49]. The larger values of these indices mean better classification performance. The experiment results are reported with the average classification accuracy of ten independent runs.

B. Classification Performance

1) Salinas Dataset: The HSI classification performance is first evaluated on the Salinas dataset. The average classification accuracies are reported in Table I. The best classification accuracies are highlighted in boldface letters. Overall, the proposed GoLoG consistently outperforms the other methods in terms of three important indices (i.e., OA, AA, and κ) under 0.5% of training samples ratio. In addition, the GoLoG also obtains small standard deviations. This illustrates that the GoLoG has a more reasonable network framework and also adaptively learns an informative sparse and low-rank graph structure, which effectively improves classification accuracy. In addition, the proposed GoLoG outperforms other graph structure learning-based GCN methods, such as CEGCN and MDGCN. One probable explanation is that the compared methods simply learn the graph structure through neural networks or similarity learning which cannot maintain the sparse and low-rank properties of the graph, while the proposed GoLoG learns a refined sparse and low-rank graph structure, which appropriately guides the process of aggregating neighborhood information and further boosts classification accuracy. Furthermore, GoLoG and the variant $\text{GoLoG}_{\text{base}}$ significantly surpass other GCN-based methods that do not differentiate the global stage and the local stage. A possible reason for this is that the GoLoG possesses a more reasonable global-to-local network framework, which not only captures long-range spatial contextual information in the early stage, but also emphasizes the importance of local individualized information for the HSI classification task. Overall, GCN-based methods outperform CNN-based methods on the Salinas dataset. This is reasonable because the Salinas dataset comprises many homogeneous regions, thus the GCN which has a powerful long-range modeling ability can make sense of this and aggregate the information from hyperspectral pixels that are far away in space.

Fig. 2 illustrates the visual classification maps of all the methods on the Salinas dataset. One characteristic of the Salinas dataset is that it contains a large number of homogeneous regions of different sizes, in particular, several large ones. It can be seen that the visual results produced by CNN-based methods are extremely poor and contain a lot of noisy labels, while GCN-based approaches clearly work better in these homogeneous regions. This validates that the GCN has strong ability to capture long-range dependencies and can aggregate the spatial information of HSI better, which is important to maintain the smoothness of the homogeneous regions. More importantly, it is obvious that the classification map produced by GoLoG is significantly more similar to the ground truth. Specifically, GoLoG outperforms other GCN-based methods, such as CEGCN and MDGCN, in some small homogeneous regions. This is because the GoLoG can correctly exploit long-range spatial information and the local spatial information using a global-to-local network framework. Moreover, the graph structure learning strategy guided by low-rank and sparse properties also enhances the visual result by adaptively capturing underlying graph structure of HSI. To sum up, these visual comparisons verify the superiority of GoLoG on the Salinas dataset.

2) Indian Pines Dataset: Table II and Fig. 3 present the classification accuracies and the visual classification maps of all the comparison methods on this dataset, respectively. As shown in Fig. 3, the Indian Pines dataset contains not only

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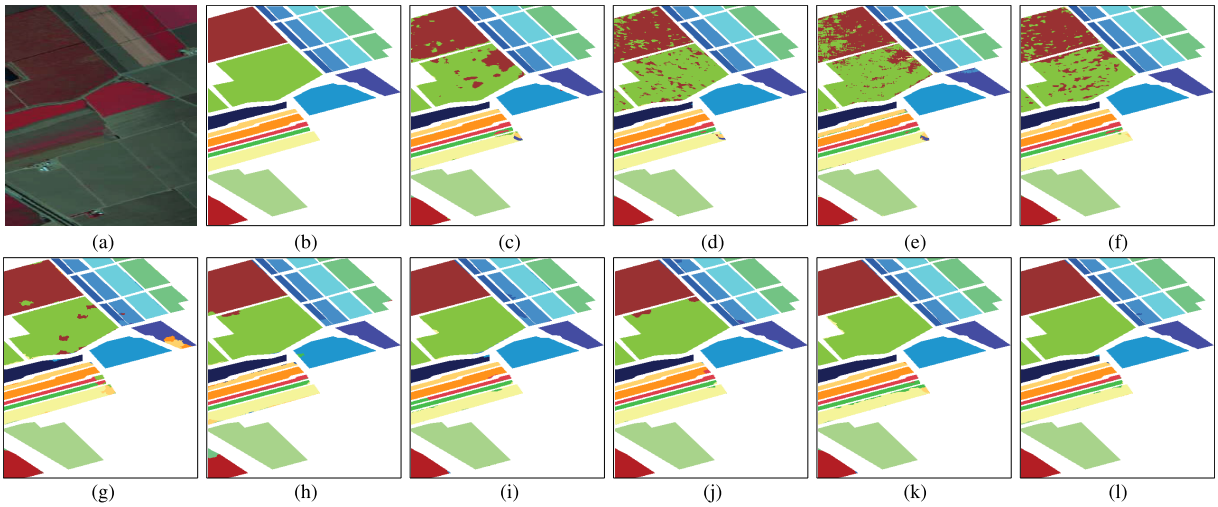


Fig. 2. Classification maps of the compared methods on the Salinas dataset. (a) False color image. (b) Ground truth. (c) 2DCNN. (d) SSRN. (e) LMAFN. (f) FGN. (g) GCN. (h) MDGCN. (i) CEGCN. (j) AMGCFN. (k) GoLoGbase. (l) GoLoG.

TABLE II

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS ON THE INDIAN PINES DATASET

Class	train	test	2D-CNN	SSRN	LMAFN	FGN	GCN	MDGCN	CEGCN	AMGCFN	$\text{GoLoG}_{\text{base}}$	GoLoG
1	3	43	87.04	89.41	95.74	81.40	95.12	90.70	96.35	94.42	95.58	**97.67**
2	72	1356	93.97	**97.33**	96.64	95.37	95.13	96.00	95.40	86.59	95.91	95.78
3	42	788	95.98	96.11	96.13	95.66	91.68	92.42	91.84	88.65	**97.04**	96.73
4	12	225	94.67	94.42	90.89	89.78	88.44	86.44	90.35	90.09	94.80	**100.00**
5	25	458	94.23	94.95	**95.82**	95.81	93.54	92.14	94.35	91.88	94.52	95.80
6	37	693	99.05	99.17	99.45	**99.77**	97.10	97.58	97.49	96.64	98.17	96.97
7	2	26	99.45	94.44	99.36	83.85	100.00	94.23	99.45	10.00	93.08	**100.00**
8	24	454	99.87	99.41	99.74	99.69	99.89	95.76	99.59	99.71	99.96	**100.00**
9	1	19	91.73	85.38	91.23	68.42	**98.95**	78.95	84.21	27.37	80.00	94.74
10	49	923	94.51	95.39	94.31	94.73	91.20	89.65	90.74	81.50	92.71	**96.64**
11	123	2332	93.74	96.35	96.12	95.61	95.27	96.25	96.52	95.58	98.73	**99.01**
12	30	563	95.00	96.51	93.37	95.88	94.37	94.14	94.62	88.93	97.18	**98.76**
13	11	194	**99.93**	99.31	99.66	98.76	98.51	98.84	98.01	97.53	98.40	99.48
14	64	1201	98.87	98.39	98.42	97.79	99.18	99.00	99.61	99.70	99.82	**100.00**
15	20	366	91.45	97.33	93.21	95.25	94.81	96.99	95.16	**100.00**	98.50	99.18
16	5	88	97.89	98.36	99.43	98.41	95.23	99.15	99.84	**99.89**	97.61	99.15
OA(%)	-	-	95.50±0.40	96.93±0.42	96.41±0.45	96.07±0.14	95.18±0.58	95.19±0.45	95.67±0.69	92.40±0.41	97.31±0.28	**98.04±0.21**
AA(%)	-	-	95.46±1.04	95.77±1.86	96.22±0.86	92.89±1.96	95.52±0.97	93.64±2.09	95.22±2.05	84.28±2.26	95.75±1.58	**98.12±0.88**
$\kappa \times 100$	-	-	94.87±0.46	96.50±0.48	95.90±0.51	95.52±0.15	94.50±0.66	94.52±0.52	95.06±0.79	91.33±0.46	96.93±0.32	**97.76±0.24**

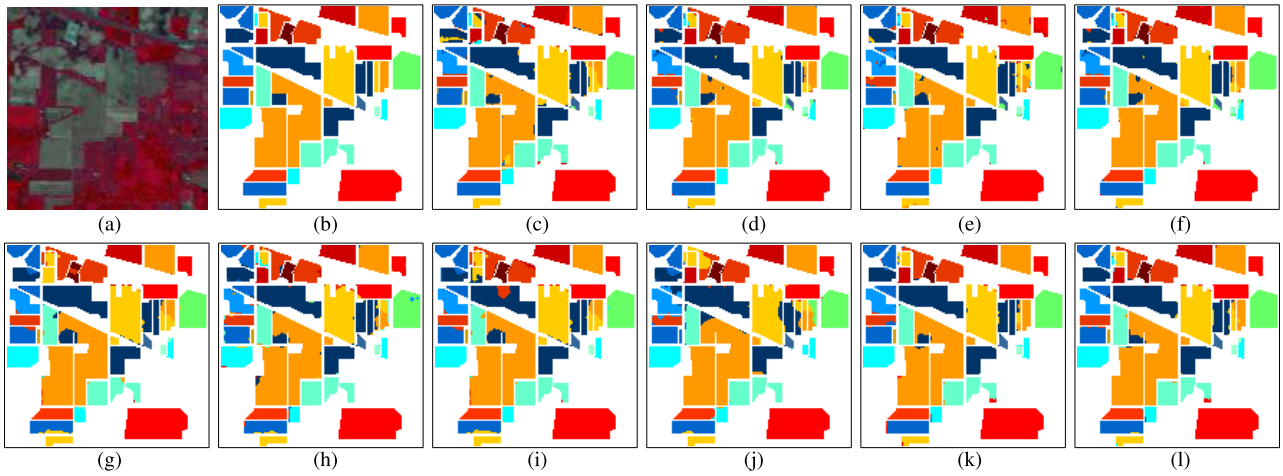


Fig. 3. Classification maps of the compared methods on the Indian Pines dataset. (a) False color image. (b) Ground truth. (c) 2DCNN. (d) SSRN. (e) LMAFN. (f) FGN. (g) GCN. (h) MDGCN. (i) CEGCN. (j) AMGCFN. (k) GoLoGbase. (l) GoLoG.

many homogeneous regions but also lots of detailed spatial structures, which makes classification challenging. Furthermore, the imbalanced labeled sample problem of this dataset further increases classification difficulty.

Table II shows that most of the compared methods have lower OA, AA, and κ values, and also perform badly in most of the classes except SSRN, LMAFN, and the proposed $\text{GoLoG}_{\text{base}}$ and GoLoG. One possible reason of this is that these CNN-based methods do not exploit long-range spatial information while GCN-based methods are not able to model local spatial information well, and both result in poor accuracy. On the contrary, the global-to-local network framework of the

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TABLE III
CLASSIFICATION ACCURACIES OF THE COMPARED METHODS ON THE PAVIA UNIVERSITY DATASET

Class	train	test	2D-CNN	SSRN	LMAFN	FGN	GCN	MDGCN	CEGCN	AMGCFN	$\text{GoLoG}_{\text{base}}$	GoLoG
1	34	6597	97.66	98.82	96.05	98.22	90.25	90.66	94.27	96.68	**98.85**	98.62
2	94	18555	98.97	99.66	99.55	99.65	99.16	98.84	99.68	99.51	99.47	**99.89**
3	11	2088	86.39	88.42	81.87	87.27	97.57	87.09	94.44	96.84	97.39	**99.72**
4	16	3048	95.45	**96.38**	95.73	95.84	68.08	83.94	83.90	90.44	93.90	95.51
5	7	1338	99.96	100.00	99.70	99.96	99.85	97.63	98.97	99.90	99.95	**100.00**
6	26	5003	91.17	97.63	96.84	98.01	97.75	97.66	98.42	**99.95**	98.82	99.90
7	7	1323	86.90	98.73	88.89	98.06	96.30	94.29	96.36	98.32	95.99	**100.00**
8	19	3663	93.40	95.47	94.62	94.53	90.85	91.63	93.17	98.32	97.80	**98.43**
9	5	942	96.76	99.81	98.25	**99.94**	48.85	79.96	60.58	76.88	93.03	97.53
OA(%)	-	-	96.11±0.55	98.13±0.50	96.77±0.78	97.88±0.28	93.41±0.22	94.57±0.79	95.76±0.59	97.72±0.58	98.42±0.39	**99.20±0.09**
AA(%)	-	-	94.07±1.44	97.21±0.74	94.61±1.80	96.83±0.53	87.63±1.08	91.30±1.15	91.09±1.80	95.21±1.33	97.24±0.85	**98.84±0.22**
$\kappa \times 100$	-	-	94.82±0.73	97.52±0.66	95.71±1.05	97.19±0.37	91.24±0.30	92.80±1.04	94.35±0.79	96.97±0.77	97.90±0.51	**98.94±0.12**

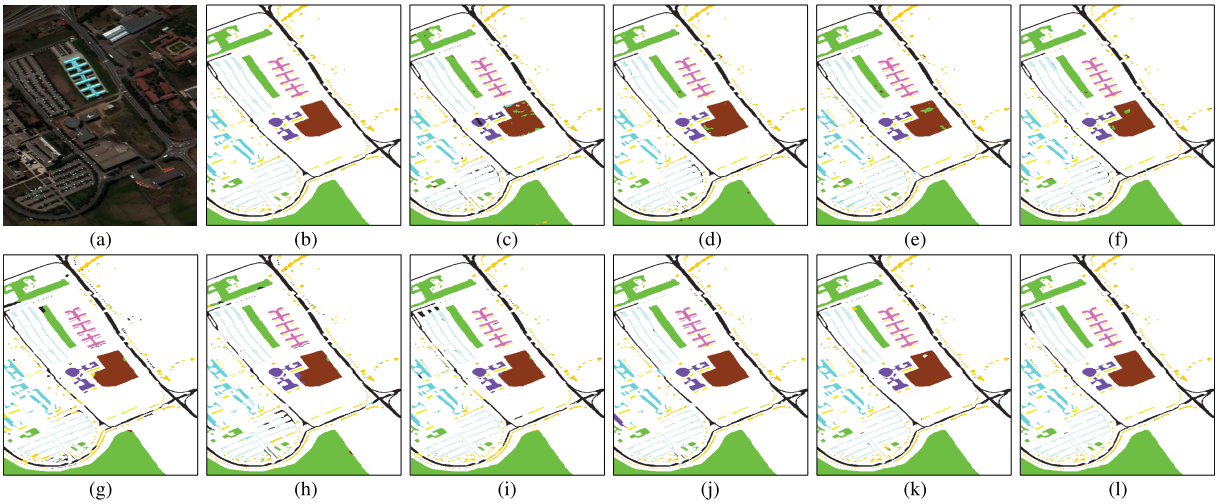


Fig. 4. Classification maps of the compared methods on the Pavia University dataset. (a) False color image. (b) Ground truth. (c) 2DCNN. (d) SSRN. (e) LMAFN. (f) FGN. (g) GCN. (h) MDGCN. (i) CEGCN. (j) AMGCFN. (k) GoLoGbase. (l) GoLoG.

GoLoG makes it possible to properly utilize the long-range spatial contextual information and also use local individualized information well.

As shown in Fig. 3, GCN-based methods are not clearly superior compared with CNN-based methods, which is different from the results in the Salinas dataset. It can be found that the CNN-based methods perform better in preserving detailed spatial structures, while GCN-based methods maintain smoothness in the homogeneous regions. This is reasonable because CNN has a powerful capability of modeling local spatial information, while GCN is good at exploiting long-range spatial information. These analyses indicate the different importance of local information and global information to HSI classification task.

3) Pavia University Dataset: Table III and Fig. 4, respectively, present the classification accuracies and the classification maps of all comparison methods on the Pavia University dataset. Table III shows that the GoLoG achieves the best OA, AA, and κ with the smallest standard deviations. The GoLoG surpasses the GCN and the variant $\text{GoLoG}_{\text{base}}$, which again reflects the importance of learning the appropriate graph structure of HSI. In addition, the GoLoG performs better than the other graph structure learning-based GCN methods. There are two possible reasons for this. First, the GoLoG applies a global-to-local network framework which makes properly make full use of the global and local spatial information in HSI. Second, the graph structure learning strategy of the GoLoG obtains the global graph which satisfies the intrinsic graph properties. Additionally, a comparison of the $\text{GoLoG}_{\text{base}}$ and the other GCN-based methods illustrates that the global-to-local network framework significantly refines the utilization of the global and local spatial information in HSI.

It can be concluded from Fig. 4 that the CNN-based methods and GCN-based methods have different advantages, namely the CNN-based methods produce satisfactory results in the local region, while the GCN-based methods can greatly preserve smoothness in the homogeneous regions. This is consistent with the fact that the CNN shows a powerful capability to extract local information, while the GCN performs better in exploiting global information. However, it should be emphasized that the classification map of GoLoG is the most similar to the ground truth of the compared methods, especially the homogeneous regions and local detail regions. This further verifies that GoLoG improves the utilization of global and local spatial information using the proposed global-to-local network framework. Meanwhile, benefiting from the joint optimization of the graph structure and the network parameters, GoLoG is also capable of improving the visual result and the classification accuracy.

4) Classification Under Different Training Percentages: In this section, the effects of different training percentages on classification accuracy are analyzed. The training percentage is varied from 0.3% to 0.7% with a step size of 0.1% for the Salinas and Pavia University datasets, and the training percentage is varied from 3% to 7% with a step size of 1% for the Indian Pines dataset. Fig. 5 shows that all these methods

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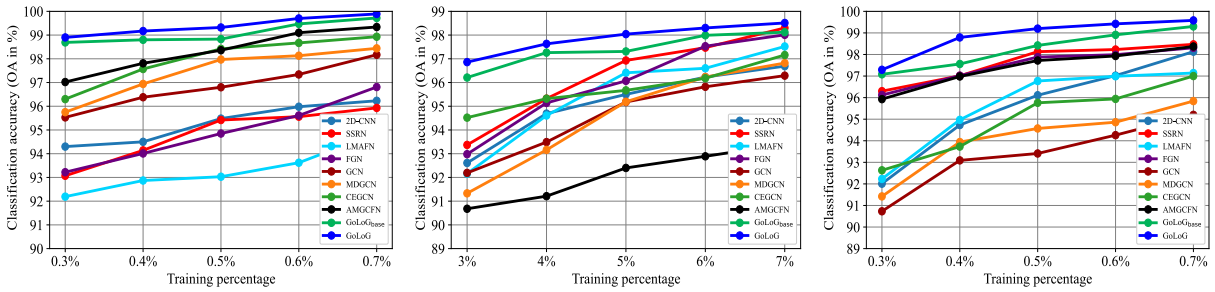


Fig. 5. Classification results of the compared methods with different training percentages on three datasets. (a) Salinas. (b) Indian Pines. (c) Pavia University.

TABLE IV
COMPUTATIONAL COMPLEXITY ON THREE DATASETS

Methods	Salinas			Indian Pines			Pavia University		
	Training time	Testing time	Parameters	Training time	Testing time	Parameters	Training time	Testing time	Parameters
GCN	21.81	0.0115	45,480	15.81	0.0107	44,960	41.99	0.0357	31,447
MDGCN	36.24	0.0146	84,912	30.19	0.0141	83,376	64.04	0.0404	43,419
CEGCN	24.63	0.0132	130,984	18.08	0.0145	128,784	52.93	0.0374	91,095
AMGCFN	30.37	0.0160	189,708	52.76	0.0130	189,448	25.49	0.0280	181,799
GoLoG$_{base}$	19.55	0.0128	143,888	18.18	0.0123	110,352	46.73	0.0454	80,521
GoLoG	74.73	0.0136	143,888	50.71	0.0135	110,352	106.19	0.0451	80,521

achieve good performance under a large training sample percentage. However, many methods fail to achieve satisfactory accuracy when the training sample percentage is relatively low. For example, the LMAFN performs very poorly under the 3% training percentage on the Indian Pines dataset, while the AMGFCN and GCN do not work well on the Salinas and Pavia University datasets under only 0.3% training percentage, respectively. On the contrary, our methods, i.e., the GoLoG and GoLoG$_{base}$, still outperform the other baselines by a large margin even under a limited training sample percentage.

5) Computational Complexity: Table IV summarizes the running time and the number of parameters of the GCN-based methods. It can be concluded that graph structure learning-based GCN methods (MDGCN, CEGCN, and GoLoG) take more time during the training stage than other GCN-based ones. This is reasonable because graph structure learning-based GCN methods need extra time to learn a refined graph, which leads to increased running time. When comparing the running time of the GoLoG and the GoLoG$_{base}$, we find that most of the training time is used to jointly optimize the graph structure and network parameters. Furthermore, the GoLoG$_{base}$ has better computational efficiency, which means that the base global-to-local network framework is efficient as it makes full use of the global contextual information and local information.

C. Ablation Study

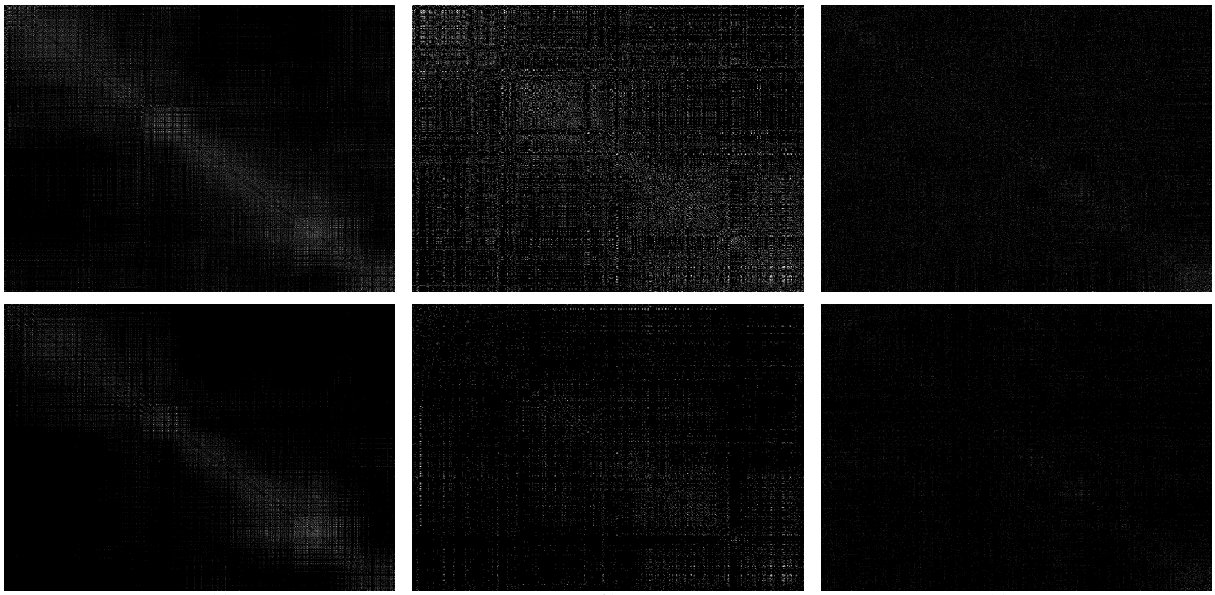
In this section, the effects of major components of the proposed GoLoG, i.e., the global-to-local framework and the sparse and low-rank graph structure learning model, are systematically evaluated over the classification performance.

1) Enhancing Discriminative Ability With the Sparse and Low-Rank Graph Structure Learning Model: To evaluate the benefit of the sparse and low-rank graph structure learning model, the counterpart model GoLoG$_{base}$ is used as the baseline, which is constructed by utilizing same global-to-local network framework but is trained without the sparse and low-rank graph structure learning. The other hyperparameters are identically set in two models. Table V presents the quantitative comparison of the GoLoG and the GoLoG$_{base}$ with the best accuracies highlighted in boldface letters. The GoLoG achieves higher classification accuracy in comparison with GoLoG$_{base}$. These results demonstrate convincingly that considering the sparsity and low-rank of the graph makes sense and clearly improves classification accuracy.

In addition to the performance comparison, we also present the visualization of the learned graph structures on the Salinas, Indian Pines, and Pavia University datasets. The visual maps are displayed in Fig. 6. Furthermore, the sparsity and low-rank of the learned graph structure are also visualized and compared with those of the original graph structure. Specifically, the sparsity is portrayed by the number of nonzero elements of the graph structure in this work, and Fig. 7 illustrates the sparsity of the graph structure before and after graph structure learning. It can be seen that the sparsity of graph structure is improved by the proposed sparse and low-rank graph structure learning model. As for the low-rank, we calculate the singular values of the graph structure and sort them from large to small, and the results are shown in Fig. 8. It can be concluded that the low-rank of the graph structure can also be enhanced by the proposed graph structure learning method.

2) Boosting Model Performance With the Global-to-Local Network Framework: We are interested in how the HSI classification task benefits from the global-to-local network framework. To do this, the GoLoG and GoLoG$_{base}$ are compared with three variants, denoted by GoLoG$_{LG}$, GoLoG$_{G}$, and GoLoG$_{L}$. Specifically, the GoLoG$_{LG}$ has the same number of layers in the local and GGCMs, and is not jointly optimized with the sparse and low-rank graph structure learning model. The GoLoG$_{G}$ contains only the GGCM and has the same number of global graph convolutional layers as the GoLoG, where the global graph is optimized by the sparse and low-rank graph structure learning model. The GoLoG$_{L}$ contains only the LGCM and has the same number of local

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(a) (b) (c)

Fig. 6. Visualization of the graph structures. The first row shows the adjacency matrixes of the original global graph structures on the Salinas, Indian Pines, and Pavia University datasets, respectively. The second row shows the adjacency matrixes of the learned global graph structures on the three datasets. (a) Salinas. (b) Indian Pines. (c) Pavia University.

TABLE V

RESULTS OF THE ABLATION STUDY ON THE SALINAS, INDIAN PINES, AND PAVIA UNIVERSITY DATASETS

	Salinas			Indian Pines			Pavia University		
	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$
GoLoG_G	96.83	97.45	96.47	96.78	92.41	96.33	96.11	92.51	94.83
GoLoG_L	98.54	98.25	98.37	96.98	94.91	96.55	97.35	96.27	96.49
GoLoG_LG	96.68	97.74	96.31	96.65	93.02	96.18	94.99	92.21	93.34
GoLoG_base	98.83	98.63	98.70	97.31	95.75	96.93	98.42	97.24	97.90
GoLoG	**99.32**	**99.02**	**99.24**	**98.04**	**98.12**	**97.76**	**99.20**	**98.84**	**98.94**

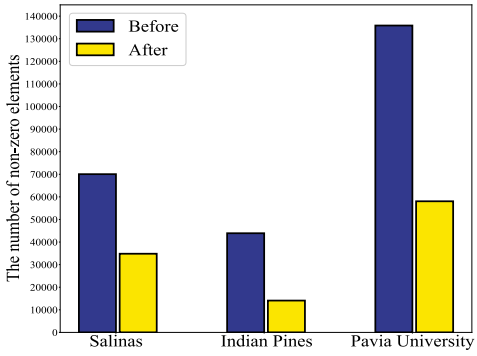


Fig. 7. Sparsity of graphs before and after graph structure learning.

graph convolutional layers as the GoLoG. Table V reports the classification performance on the three datasets, and the best accuracies are highlighted in boldface letters. Overall, the GoLoG and GoLoG_base show obvious superiority when compared with the GoLoG_LG. This validates that the global-to-local network framework helps to refine the utilization of the global and local information and further boosts classification performance.

D. Parameter Analysis

The sensitivity of key hyperparameters is explored for GoLoG in this section. There are two key hyperparameters in the sparse and low-rank graph structure learning model, i.e., α and β, which control the contributions of the sparse constraint and the low-rank constraint, respectively. The hyperparameters α and β are tuned together to find the best combination. Specifically, α and β are varied from 0 to 1 with a step size of 0.1 while the other hyperparameters remain unchanged. The accuracy changes of the GoLoG on the three datasets are illustrated in Fig. 9. For the Salinas dataset, the overall trend of performance change can be summarized as follows. The OAs demonstrate an obvious performance improvement with increasing α and β, and then decrease slightly, and finally fluctuate. The best OA is achieved when $\alpha = 0.3$ and $\beta = 0.3$ on the Salinas dataset. For the Indian Pines and Pavia University datasets, it can be seen from Fig. 9 that, with the increase of α and β, the OAs show an overall increasing trend at first and then fluctuate around a satisfactory accuracy. The best OA is achieved when $\alpha = 0.8$ and $\beta = 0.5$ on the Indian Pines dataset, while $\alpha = 0.7$ and $\beta = 0.4$ are optimal on the Pavia University dataset. To sum up, it can be concluded that the classification accuracy of the GoLoG is improved when appropriate values for α and β are chosen. This is consistent with the fact that a real-world graph possesses certain intrinsic graph properties which are important and should be maintained when learning a refined graph structure.

The hyperparameter Q in the LGCM is also very important for the GoLoG. The search range of Q is constrained in

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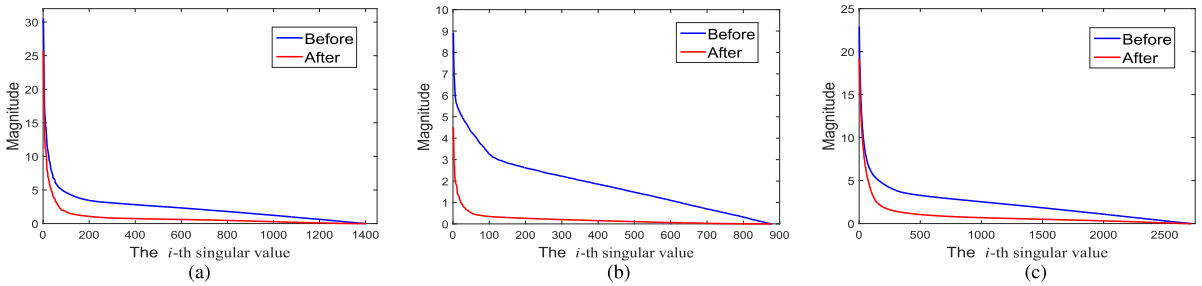


Fig. 8. Singular values of graphs before and after graph structure learning. (a) Salinas. (b) Indian Pines. (c) Pavia University.

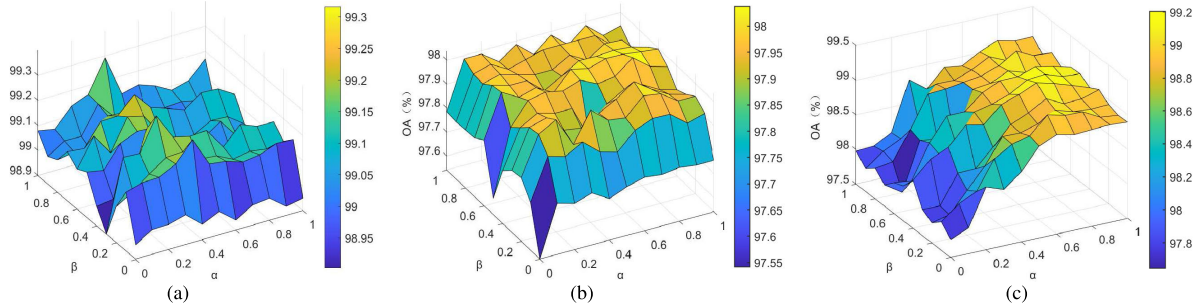


Fig. 9. Parameter analysis of α and β on three datasets. (a) Salinas. (b) Indian Pines. (c) Pavia University.

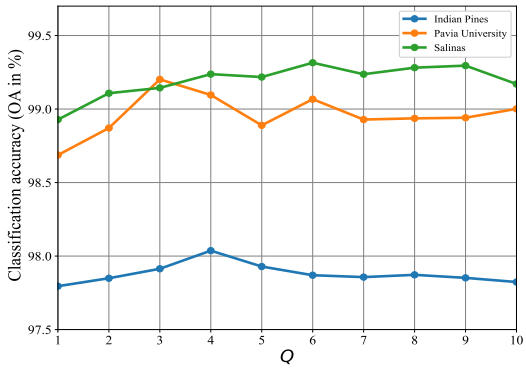


Fig. 10. Parameter analysis of Q on the three datasets.

$\{1, 2, \ldots, 10\}$. The classification accuracy of different Q is summarized in Fig. 10. It can be found that accuracy increases when Q becomes relatively larger, but the model performance slightly decreases as Q continues increasing. The best accuracy is obtained when Q is 6, 4, and 3 for the Salinas, Indian Pines, and Pavia University datasets, respectively.

V. Conclusion

This article presents a GoLoG for HSI classification, which jointly optimizes the graph structure and network parameters guided by both intrinsic graph properties and classification loss. First, through the proposed global-to-local network framework with successive global and local graph convolutional stages, the GoLoG not only exploits global contextual information but also emphasizes the individualized information of each hyperspectral pixel, which is suitable for highly accurate pixelwise HSI classification. Second, benefitting from the proposed sparse and low-rank graph structure learning model, the GoLoG significantly enhances classification accuracy with a refined global graph. Furthermore, using aid of the joint alternating update algorithm, the sparse and low-rank graph structure learning model and the global-to-local network can be unified and further cooperate closely to obtain a refined graph structure and improved classification results. Extensive experiments demonstrate the effectiveness and the superiority of the GoLoG in terms of both quantitative metrics and visual results. In future research, we plan to conceptualize and develop a segmentation network, with the intention of seamlessly integrating it with the proposed GoLoG in an end-to-end fashion. Additionally, in the realm of hyperparameter setting, we aim to investigate more advanced strategies, including but not limited to meta-learning, to facilitate the adaptive selection of optimal hyperparameters, thereby enhancing the model's performance and efficacy.

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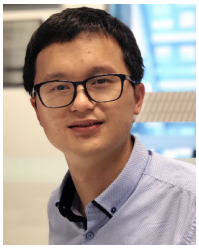
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